

Project Name: SecureStack

Platform Used: AWS EC2

Operating System: Ubuntu Server 24.04 LTS

Web Server: Apache2

Application Type: Static Website

AWS EC2 Instance Setup

The screenshot shows the AWS Management Console for the 'ap-southeast-2' region. The 'Instances' page is active, displaying a list of EC2 instances. One instance, 'SecureStack-WebServer' (ID: i-008405d1694ffdb7a), is highlighted and its details are shown in the main panel. The instance is in a 'Running' state, using the 't3.micro' instance type. The public IPv4 address is 13.238.255.68, and the public DNS name is ec2-13-238-255-68.ap-southeast-2.compute.amazonaws.com. The console also shows the instance's status, monitoring, security, networking, storage, and tags.

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
SecureStack-WebServer	i-008405d1694ffdb7a	Running	t3.micro	3/3 checks passed	ap-southeast-2c	ec2-13-238-255-68.ap-southeast-2.compute.amazonaws.com

Instance summary

- Instance ID: i-008405d1694ffdb7a
- Public IPv4 address: 13.238.255.68
- Private IPv4 addresses: 172.31.22.195
- Instance state: Running
- Public DNS: ec2-13-238-255-68.ap-southeast-2.compute.amazonaws.com

Application Deployment

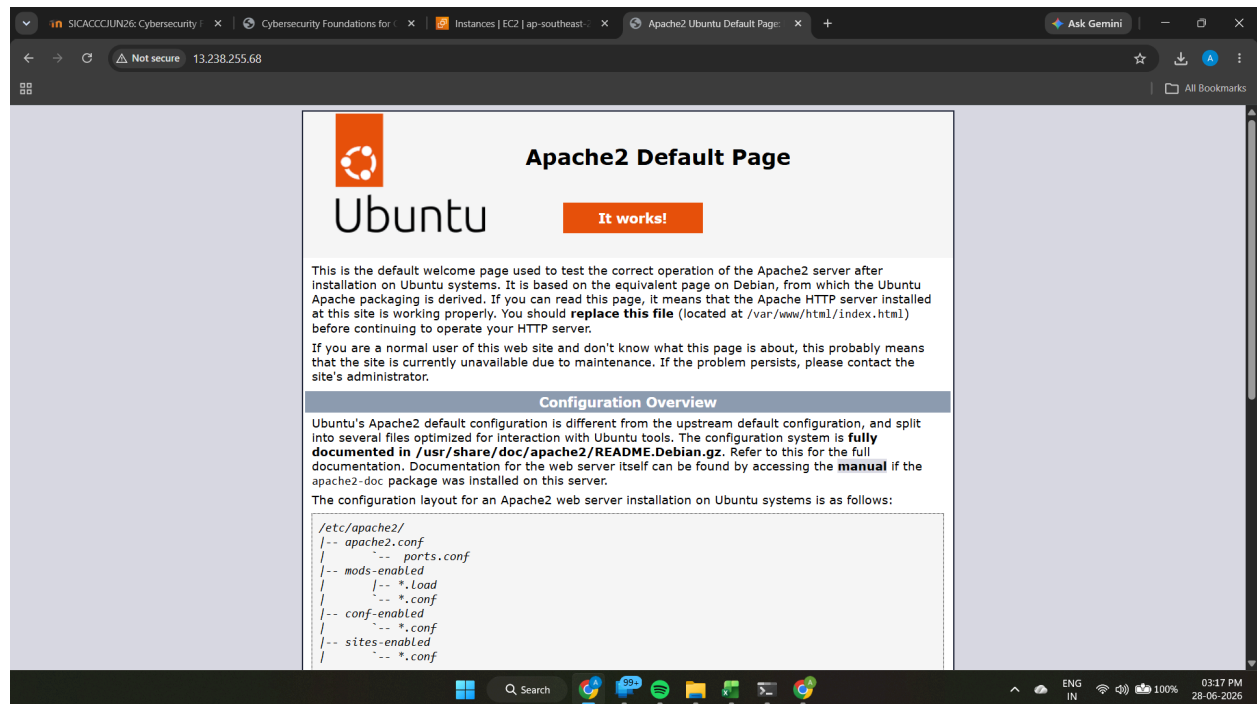
```
ubuntu@ip-172-31-22-195: /w  × + v
No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-22-195:~$ sudo apt install apache2 -y
Installing:
  apache2

Installing dependencies:
  apache2-bin      libapr1t64      libaprutil1t64
  apache2-data     libaprutil1-dbd-sqlite3  liblua5.4-0
  apache2-utils   libaprutil1-ldap  ssl-cert

Suggested packages:
  apache2-doc      | apache2-suexec-custom
  apache2-suexec-pristine  www-browser

Summary:
  Upgrading: 0, Installing: 10, Removing: 0, Not Upgrading: 0
  Download size: 2121 kB
  Space needed: 8209 kB / 4670 MB available

Get:1 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 libapr1t64 amd64 1.7.6-3 [109 kB]
Get:2 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 libaprutil1t64 amd64 1.6.3-3ubuntu3 [94.1 kB]
Get:3 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 libaprutil1-dbd-sqlite3 amd64 1.6.3-3ubuntu3 [11.5 kB]
Get:4 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 libaprutil1-ldap amd64 1.6.3-3ubuntu3 [9266 B]
Get:5 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 liblua5.4-0 amd64 5.4.8-1build1 [163 kB]
Get:6 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 apache2-bin amd64 2.4.66-2ubuntu2.2 [1358 kB]
Get:7 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 apache2-data all 2.4.66-2ubuntu2.2 [163 kB]
Get:8 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 apache2-utils amd64 2.4.66-2ubuntu2.2 [93.8 kB]
Get:9 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 apache2 amd64 2.4.66-2ubuntu2.2 [100 kB]
Get:10 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 ssl-cert all 1.1.3ubuntu2 [18.8 kB]
Fetched 2121 kB in 0s (37.5 MB/s)
Preconfiguring packages ...
Selecting previously unselected package libapr1t64:amd64.
(Reading database ... 85280 files and directories currently installed.)
Preparing to unpack .../0-libapr1t64_1.7.6-3_amd64v3.deb ...
Unpacking libapr1t64:amd64 (1.7.6-3) ...
Selecting previously unselected package libaprutil1t64:amd64.
Preparing to unpack .../1-libaprutil1t64_1.6.3-3ubuntu3_amd64v3.deb ...
Unpacking libaprutil1t64:amd64 (1.6.3-3ubuntu3) ...
Selecting previously unselected package libaprutil1-dbd-sqlite3:amd64.
Preparing to unpack .../2-libaprutil1-dbd-sqlite3_1.6.3-3ubuntu3_amd64v3.deb ...
Unpacking libaprutil1-dbd-sqlite3:amd64 (1.6.3-3ubuntu3) ...
```



Firewall Configuration

```
ubuntu@ip-172-31-22-195: /var/www/html$ sudo systemctl restart apache2
ubuntu@ip-172-31-22-195: /var/www/html$ sudo apt install ufw -y
ufw is already the newest version (0.36.2-9build1).
ufw set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0
ubuntu@ip-172-31-22-195: /var/www/html$ sudo ufw allow 22
Rules updated
Rules updated (v6)
ubuntu@ip-172-31-22-195: /var/www/html$ sudo ufw allow 80
Rules updated
Rules updated (v6)
ubuntu@ip-172-31-22-195: /var/www/html$ sudo ufw allow 443
Rules updated
Rules updated (v6)
ubuntu@ip-172-31-22-195: /var/www/html$ sudo ufw enable
Firewall is active and enabled on system startup
ubuntu@ip-172-31-22-195: /var/www/html$ sudo ufw status
Status: active

To Action From
--
22 ALLOW Anywhere
80 ALLOW Anywhere
443 ALLOW Anywhere
22 (v6) ALLOW Anywhere (v6)
80 (v6) ALLOW Anywhere (v6)
443 (v6) ALLOW Anywhere (v6)

ubuntu@ip-172-31-22-195: /var/www/html$ |
```

Purpose:

- Allows only required network traffic
- Blocks unauthorized access
- Protects the server from unnecessary exposure

Secure Authentication

By setting :

PasswordAuthentication no

In SSH Configuration, we are able to enable secure Authentication

Purpose:

- Prevents unauthorized password attacks
- Allows only authenticated key-based access

Web Protection

Apache security header added:

Purpose:

- Provides protection against clickjacking attacks
- Improves web application security

```

ubuntu@ip-172-31-22-195: /v
ubuntu@ip-172-31-22-195:/var/www/html$ sudo nano /etc/apache2/apache2.conf
ubuntu@ip-172-31-22-195:/var/www/html$ sudo systemctl restart apache2
ubuntu@ip-172-31-22-195:/var/www/html$ sudo ufw status
Status: active

To Action From
--
22 ALLOW Anywhere
80 ALLOW Anywhere
443 ALLOW Anywhere
22 (v6) ALLOW Anywhere (v6)
80 (v6) ALLOW Anywhere (v6)
443 (v6) ALLOW Anywhere (v6)

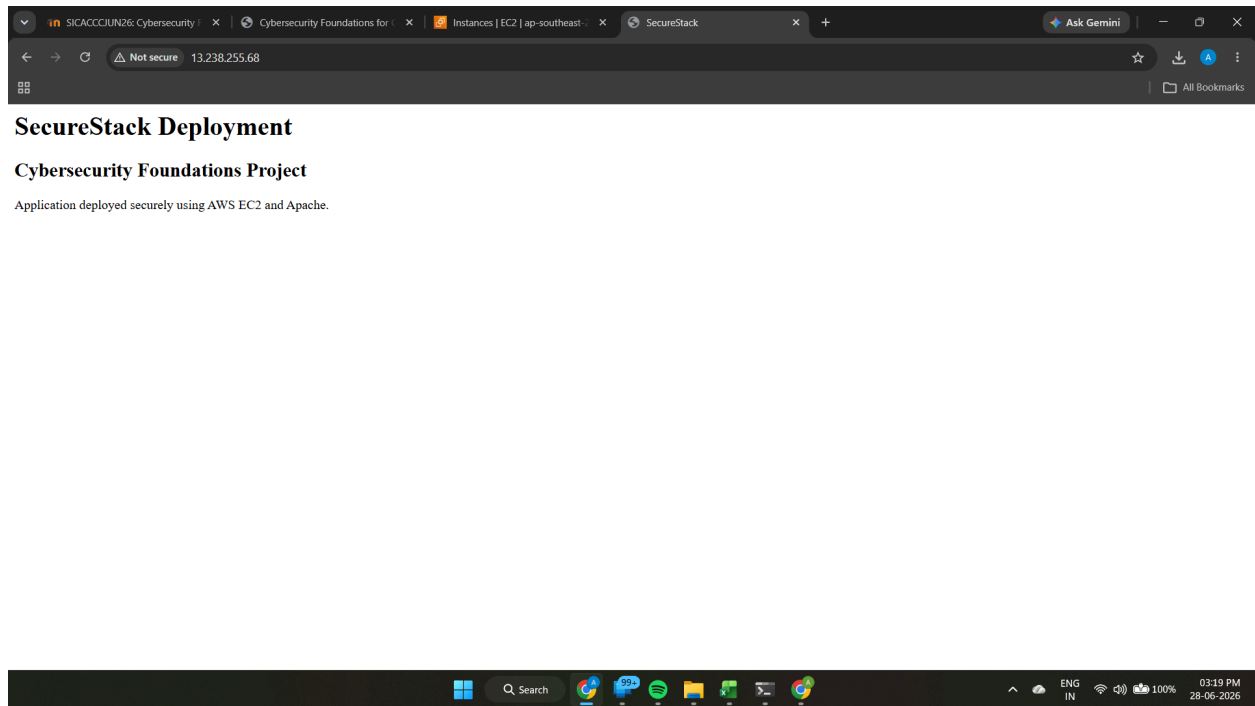
ubuntu@ip-172-31-22-195:/var/www/html$ systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-06-28 09:55:30 UTC; 21s ago
   Invocation: c5490768c8474a87b672926fda5da09
   Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 13455 (apache2)
   Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
   Tasks: 55 (limit: 627)
   Memory: 5.9M (peak: 5.9M)
   CPU: 65ms
   CGroup: /system.slice/apache2.service
           └─13455 /usr/sbin/apache2 -k start -DFOREGROUND
             └─13458 /usr/sbin/apache2 -k start -DFOREGROUND
               └─13459 /usr/sbin/apache2 -k start -DFOREGROUND

Jun 28 09:55:30 ip-172-31-22-195 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Jun 28 09:55:30 ip-172-31-22-195 systemd[1]: Started apache2.service - The Apache HTTP Server.
lines 1-17...skipping...
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-06-28 09:55:30 UTC; 21s ago
   Invocation: c5490768c8474a87b672926fda5da09
   Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 13455 (apache2)
   Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
   Tasks: 55 (limit: 627)

```

[illegible]

AS OF FINAL APPLICATION :



PUBLIC IP : <http://13.238.255.68/>